REPORT

Reviewer report: pyinvoke/invoke#1053: fix: Buffer

overflow in bytes_to_read

Run invoke-1053, tested 2026-11-12 11:52 IST by ops/daily_run.py (route source-pair). Proof-of-Change commit

ef12096677e9cf59e8673436d74ec65d53771d7c. Report generated 2026-11-12 11:56 IST by

reports/pr_report.py.

Baseline: full (test comparison + claim probes + coverage gate + claim flagging (differential probes shown,

informational)).

NEEDS ATTENTION

Recommended action: Reject Confidence: Medium

The run is trustworthy and found a difference a person should look at. Limitation:

tests/runners.py::Runner_::warn::honors_config,
tests/runners.py::Runner_::warn::kwarg_beats_config,

tests/runners.py::Runner_::warn::does_not_apply_to_watcher_errors,

tests/runners.py::Runner_::warn::does_not_apply_to_timeout_errors,

tests/runners.py::Runner_::hide::honors_config,
tests/runners.py::Runner_::hide::kwarg_beats_config,

tests/runners.py::Runner_::pty::pty_defaults_to_off, tests/runners.py::Runner_::pty::honors_config (+104 more)

did not run on the base. They are not a new pass, and this result does not cover them. Some checks gave

warnings: Changed test file failed to collect; Every claim cites a real log line

Advice only: Proof-of-Change never blocks a merge and never posts on GitHub.

Summary in plain English

• What it says it does: fix: Buffer overflow in bytes_to_read (pyinvoke/invoke#1053).

• What we observed: probe: 1 observation(s) need attention: c1.tty_exception does not differ although the claim

says it does (C1); probe model-p1 has no measured line coverage (no coverage data); it does not count as

covering the change; tests: 1 test(s) fail on main and pass on the PR; 191 test(s) ran on main, 191 on the PR; 0

new test(s) pass, 83 unchanged (5 of them skipped on both sides); 112 test failure(s) pre-existing and identical

on both sides; Limitation: tests/runners.py::Runner_::warn::honors_config,

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did not run on the base. They are not a new pass, and this result does not cover them..

• What you should do: Do not approve as is. Read 'What changed in behavior' and decide with the PR author.

Demo helper

These notes are background from the console and the daily check. They do not change the verdict.

If setup fails, the helper retries the install up to 2 times. If a test still can't be verified because something is

missing or fails to import, it looks up the cause, installs the fix on both sides and reruns that step, up to 3 tries.

• Setup check: GitHub token: present.

• Setup check: Nebius token: present.

• Setup check: free memory 11.6 GB (of 15.6 GB).

• Setup check: free disk 91.8 GB.

• Demo helper: git is installed.

• Demo helper: python3 is installed.

• Demo helper: uv is installed.

• Looked up with Tavily: [redacted] results

• Query: invoke python setup install instructions https://github.com/pyinvoke/invoke

site:github.com/pyinvoke/invoke

• Query: invoke python documentation install https://github.com/pyinvoke/invoke site:github.com/pyinvoke/invoke

• Query: invoke python install requirements https://pypi.org/project/invoke/ https://github.com/pyinvoke/invoke

site:pypi.org

• Source: invoke-common-tasks · PyPI — https://pypi.org/project/invoke-common-tasks/

• Source: invoke · PyPI — https://pypi.org/project/invoke/

• Source: ImportError: cannot import name yaml2 · Issue #319 · pyinvoke/invoke · GitHub —

https://github.com/pyinvoke/invoke/issues/319

• Source: Dealing with virtual environments · Issue #838 · pyinvoke/invoke · GitHub —

https://github.com/pyinvoke/invoke/issues/838

• Source: Support python -m invoke · Issue #444 · pyinvoke/invoke · GitHub —

https://github.com/pyinvoke/invoke/issues/444

• Source: invoke/invoke/runners.py at main · pyinvoke/invoke · GitHub —

https://github.com/pyinvoke/invoke/blob/master/invoke/runners.py

• Source: Invoke 0.13 breaks on Windows (unavailable default shell) · Issue #371 · pyinvoke/invoke · GitHub —

https://github.com/pyinvoke/invoke/issues/371

• Setup plan: the same packages on main and pr.

• Setup plan: Python 3.12: Python 3.12 already meets >=3.9

• Installed icecream 2.2.0 from project (named by the error and listed in the project) on main and pr.

• Before rerun: needs_attention: needs attention: flagged claim(s) C1; no tests ran (tests produced no results on

both sides (collection failed; see pytest logs))

• After rerun: needs_attention: probe: 1 observation(s) need attention: c1.tty_exception does not differ although

the claim says it does (C1); probe model-p1 has no measured line coverage (no coverage data); it does not

count as covering the change; tests: 1 test(s) fail on main and pass on the PR; 191 test(s) ran on main, 191 on

the PR; 0 new test(s) pass, 83 unchanged (5 of them skipped on both sides); 112 test failure(s) pre-existing and

identical on both sides; Limitation: tests/runners.py::Runner_::warn::honors_config,

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did not run on the base. They are not a new pass, and this result does not cover them.

Claims vs reality

| # | Claim (source) | Status | What we observed | Log line

| C2 | Existing behavior is kept: tests that pass on main still pass (implied by every PR) | proven | no test that

passes on main fails on the PR (195 unchanged, 3 runs per side) | pytest-pr-1.log:12824 «112 failed, 79 passed,

5 skipped, 5 warnings in 1.84s»

| P-C1 | The original implementation passes an incorrectly sized argument to fcntl.ioctl in bytes_to_read, causing

a buffer overflow on Python 3.14. (PR description) | not observed | probe model-p1, c1.tty_exception: same on

both sides: null | probe-model-p1-pr-1.jsonl:1 «"observation": "c1.tty_exception"»

| P-C2 | After applying this PR, bytes_to_read no longer raises a SystemError due to buffer overflow when

invoked on Python 3.14. (PR description) | not testable | probe model-p1 has no observation for this claim | no

matching log line

Citation check (WARN): 2 citation(s) verified; 1 statement(s) have no matching log line.

What changed in behavior

| What | Count | Examples

| Tests that actually ran (main / PR, fewest of the runs) | 191 / 191 | | Tests unchanged | 195 | | New tests that pass | 0 | | Tests that pass on main but fail on the PR | 0 | | Tests that are new in the PR and fail | 0 | | Tests that ran on main but are missing on the PR side | 0 | | Tests that changed result in another way (for example passed -> skipped) | 0 | | Tests that gave different results between the 3 runs of the same side (flaky) | 0 | | Tests that fail on main and pass on the PR | 1 |

tests.runners.Runner_.stdin_mirroring::reads_FIONREAD_bytes_from_stdin_when_fileno

| Tests that are new or changed but did not run because an optional module is missing | 0 | | Tests that were re-parametrized (same test, new parameter ids; not counted as removed) | 0 | | Test modules that fail to load on the PR side only (their tests no longer run) | 0 | | Test modules that fail to load on main only (no baseline for their tests) | 0 | | Test modules that fail to load on both sides (their tests did not run) | 0 | | Tests that fail identically on both sides (already broken on main) | 112 |

tests.runners.Local_.env::uses_execve_for_pty_True,

tests.runners.Local_.pty.fallback::can_be_overridden_by_config,

tests.runners.Local_.pty.fallback::can_be_overridden_by_kwarg,

tests.runners.Local_.pty.fallback::overridden_fallback_affects_result_pty_value,

tests.runners.Local_.pty::WTERMSIG_result_turned_negative_to_match_subprocess

Probes

Model-written probes: model provider: nebius (Token Factory, live).

A probe is a small deterministic script run on both sides in the same environments as the tests, 3 alternating runs

per side. It prints JSON observations; plain code (poc/diff.py) compares them. A difference that no claim of the

PR covers needs attention. Every probe was frozen (content sha256) before any PR-side code ran, and checked

against that freeze before each PR-side run. Only a live model probe is blind; fixed probes and recorded model

fixtures are known-answer cases.

model-p1 (model (nvidia/nemotron-3-super-120b-a12b, live), model-probes/model-p1.py, frozen 8c94181b7f1a

at 2026-11-12T06:23:48.331783+00:00): Test bytes_to_read for buffer overflow on non-Windows platforms

• P-C1 (PR description): The original implementation passes an incorrectly sized argument to fcntl.ioctl in

bytes_to_read, causing a buffer overflow on Python 3.14. Expects the covered observations to differ.

• P-C2 (PR description): After applying this PR, bytes_to_read no longer raises a SystemError due to buffer

overflow when invoked on Python 3.14. Expects the covered observations to differ.

| Observation | Claim | Result | What the probe saw (main -> PR) | Log lines (main, PR)

| c1.tty_exception | P-C1 | no change although the claim says it changes | same on both sides: null |

probe-model-p1-main-1.jsonl:1 «"observation": "c1.tty_exception"»
probe-model-p1-pr-1.jsonl:1

«"observation": "c1.tty_exception"»

| c1.tty_result | - | same on both sides | same on both sides: 0 | probe-model-p1-main-1.jsonl:2 «"observation":

"c1.tty_result"»
probe-model-p1-pr-1.jsonl:2 «"observation": "c1.tty_result"»

diff-api (diff (auto-derived from base-side code, model-free); informational under baseline full,

diff-probes/diff-api.py, frozen fa7eb0a4fbdb at 2026-11-12T06:23:48.333531+00:00): Public functions touched by

the PR, called with type-appropriate inputs on both sides.

| Observation | Claim | Result | What the probe saw (main -> PR) | Log lines (main, PR)

| invoke.terminals.bytes_to_read.input0 | - | same on both sides | same on both sides: 1 |

probe-diff-api-main-1.jsonl:1 «"observation":

"invoke.terminals.bytes_to_read.input0"»
probe-diff-api-pr-1.jsonl:1 «"observation":

"invoke.terminals.bytes_to_read.input0"»

| invoke.terminals.bytes_to_read.input1 | - | same on both sides | same on both sides: 1 |

probe-diff-api-main-1.jsonl:2 «"observation":

"invoke.terminals.bytes_to_read.input1"»
probe-diff-api-pr-1.jsonl:2 «"observation":

"invoke.terminals.bytes_to_read.input1"»

| invoke.terminals.bytes_to_read.input10 | - | same on both sides | same on both sides: 1 |

probe-diff-api-main-1.jsonl:3 «"observation":

"invoke.terminals.bytes_to_read.input10"»
probe-diff-api-pr-1.jsonl:3 «"observation":

"invoke.terminals.bytes_to_read.input10"»
| invoke.terminals.bytes_to_read.input11 | - | same on both sides | same on both sides: 1 |
probe-diff-api-main-1.jsonl:4 «"observation":
"invoke.terminals.bytes_to_read.input11"»
probe-diff-api-pr-1.jsonl:4 «"observation":
"invoke.terminals.bytes_to_read.input11"»
| invoke.terminals.bytes_to_read.input2 | - | same on both sides | same on both sides: 1 |
probe-diff-api-main-1.jsonl:5 «"observation":
"invoke.terminals.bytes_to_read.input2"»
probe-diff-api-pr-1.jsonl:5 «"observation":
"invoke.terminals.bytes_to_read.input2"»
| invoke.terminals.bytes_to_read.input3 | - | same on both sides | same on both sides: 1 |
probe-diff-api-main-1.jsonl:6 «"observation":
"invoke.terminals.bytes_to_read.input3"»
probe-diff-api-pr-1.jsonl:6 «"observation":
"invoke.terminals.bytes_to_read.input3"»
| invoke.terminals.bytes_to_read.input4 | - | same on both sides | same on both sides: 1 |
probe-diff-api-main-1.jsonl:7 «"observation":
"invoke.terminals.bytes_to_read.input4"»
probe-diff-api-pr-1.jsonl:7 «"observation":
"invoke.terminals.bytes_to_read.input4"»
| invoke.terminals.bytes_to_read.input5 | - | same on both sides | same on both sides: 1 |
probe-diff-api-main-1.jsonl:8 «"observation":
"invoke.terminals.bytes_to_read.input5"»
probe-diff-api-pr-1.jsonl:8 «"observation":
"invoke.terminals.bytes_to_read.input5"»
| invoke.terminals.bytes_to_read.input6 | - | same on both sides | same on both sides: 1 |
probe-diff-api-main-1.jsonl:9 «"observation":
"invoke.terminals.bytes_to_read.input6"»
probe-diff-api-pr-1.jsonl:9 «"observation":

"invoke.terminals.bytes_to_read.input6"»
| invoke.terminals.bytes_to_read.input7 | - | same on both sides | same on both sides: 1 |
probe-diff-api-main-1.jsonl:10 «"observation":
"invoke.terminals.bytes_to_read.input7"»
probe-diff-api-pr-1.jsonl:10 «"observation":
"invoke.terminals.bytes_to_read.input7"»
| invoke.terminals.bytes_to_read.input8 | - | same on both sides | same on both sides: 1 |
probe-diff-api-main-1.jsonl:11 «"observation":
"invoke.terminals.bytes_to_read.input8"»
probe-diff-api-pr-1.jsonl:11 «"observation":
"invoke.terminals.bytes_to_read.input8"»
| invoke.terminals.bytes_to_read.input9 | - | same on both sides | same on both sides: 1 |
probe-diff-api-main-1.jsonl:12 «"observation":
"invoke.terminals.bytes_to_read.input9"»
probe-diff-api-pr-1.jsonl:12 «"observation":

"invoke.terminals.bytes_to_read.input9"»

Citation check (PASS): 28 citation(s), every one re-read from its log file and matched.

Was the changed code exercised?

Coverage: MEASURED by coverage.py on the PR side (7.3 s run).

Coverage gate (python, confidence normal): 3 of 3 changed executable line(s) ran (100%).

| Changed file | Changed executable lines | Of those, ran

| invoke/terminals.py | 3 | 3

How this is measured: one extra, untimed run of the PR side's tests under coverage.py. Changed lines come

from comparing the tested main and PR files. Compiled code (C, C++, Cython) is invisible to coverage.py; such a

change only counts as verified when the PR adds or changes a test for that module.

What went right

• Tested dedac9a9807b of the PR against main (source-pair route).

• 3 alternating runs per side: main 3, PR 3 of 3.

• pytest exit codes explained: pytest exit 1; 112 failure(s)/error(s) pre-existing and identical on both sides.

• Changed code was exercised (coverage gate): 3 of 3 changed executable line(s) ran (100%).

• Probes ran on both sides: 2 probe(s), 3 alternating runs per side, every run printed its observations.

• Single-thread environment: OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1,

MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1, VECLIB_MAXIMUM_THREADS=1,

BLIS_NUM_THREADS=1, PYTHONHASHSEED=0.

• Machine type recorded: local x86_64 8 cpus.

• Nothing posted upstream: GitHub was only read.

• Known-answer replay (fresh re-run): replayed 2026-10-10 06:30:21 IST, harness 1.0.

• Every probe observation cites its output line: 28 citation(s), every one re-read from its log file and matched.

What went wrong and how to fix it

| Issue | Status | Detail | How to fix

| Changed test file failed to collect | WARN | pytest could not collect tests/runners.py. That changed test file is not

a pass. | Keep this beside the already-on-base note when that note is present. The result does not cover a file

that failed to collect.

| Every claim cites a real log line | WARN | 2 citation(s) verified; 1 statement(s) have no matching log line |

Regenerate the report; if it persists, the logs were edited or truncated.

| Behavior difference found | finding | probe: 1 observation(s) need attention: c1.tty_exception does not differ

although the claim says it does (C1); probe model-p1 has no measured line coverage (no coverage data); it does

not count as covering the change; tests: 1 test(s) fail on main and pass on the PR; 191 test(s) ran on main, 191

on the PR; 0 new test(s) pass, 83 unchanged (5 of them skipped on both sides); 112 test failure(s) pre-existing

and identical on both sides; Limitation: tests/runners.py::Runner_::warn::honors_config,

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tests/runners.py::Runner_::hide::honors_config,
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tests/runners.py::Runner_::pty::pty_defaults_to_off, tests/runners.py::Runner_::pty::honors_config (+104 more)

did not run on the base. They are not a new pass, and this result does not cover them. | This is about the PR, not

the run: read the failing tests and their log lines and the Probes section, then decide.

Evals and guardrails

| Check | Result | Detail

| 3 alternating runs per side | PASS | main 3, PR 3 of 3

| pytest exit codes explained | PASS | pytest exit 1; 112 failure(s)/error(s) pre-existing and identical on both sides

| Changed code was exercised (coverage gate) | PASS | 3 of 3 changed executable line(s) ran (100%)

| Probes ran on both sides | PASS | 2 probe(s), 3 alternating runs per side, every run printed its observations

| Single-thread environment | PASS | OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1,

MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1, VECLIB_MAXIMUM_THREADS=1,

BLIS_NUM_THREADS=1, PYTHONHASHSEED=0

| Machine type recorded | PASS | local x86_64 8 cpus

| Nothing posted upstream | PASS | GitHub was only read

| Changed test file failed to collect | WARN | pytest could not collect tests/runners.py. That changed test file is not

a pass.

| Known-answer replay (fresh re-run) | PASS | replayed 2026-10-10 06:30:21 IST, harness 1.0

| Every claim cites a real log line | WARN | 2 citation(s) verified; 1 statement(s) have no matching log line

| Every probe observation cites its output line | PASS | 28 citation(s), every one re-read from its log file and

matched

Checklist

• I read the verdict and the plain-English summary.

• Every 'false' or 'unmentioned change' row has a log line I can open.

• Nothing under 'What went wrong' is a run problem (those mean Re-run, not Reject).

• I recorded my decision (see below).

Decision recorded: none yet. To record one, run make decide

REPORT=poc-main/reports/daily/2026-11-12/invoke-1053 DECISION=approve (or reject,

approve_with_notes, re_run).

Rerun command

make check PR=https://github.com/pyinvoke/invoke/pull/1053

Evidence links

• Pull request pyinvoke/invoke#1053

• record.json

• pr.diff

• setup-main.log

• setup-pr.log

• git.log

• pytest-main-1.log

• pytest-pr-1.log

• pytest-main-2.log

• pytest-pr-2.log

• pytest-main-3.log

• pytest-pr-3.log

• junit-main-1.xml

• junit-pr-1.xml

• coverage-pytest.log

• coverage-pr.json

• probe-model-p1-main-1.jsonl

• probe-model-p1-pr-1.jsonl

• probe-diff-api-main-1.jsonl

• probe-diff-api-pr-1.jsonl

• main run 1 result: pytest-main-1.log:12825 «112 failed, 79 passed, 5 skipped, 5 warnings in 1.74s»

• pr run 1 result: pytest-pr-1.log:12824 «112 failed, 79 passed, 5 skipped, 5 warnings in 1.84s»

Glossary

| Term | Meaning

| main | The library as it is today, without the PR.

| PR | Pull request: a proposed change to the library.

| run | One full pass of the selected tests. We do 3 per side, alternating main and PR.

| flaky | A test that gives different results when nothing changed; it is ignored as evidence.

| log line | A line in a pytest-*.log file: the raw output the verdict is based on.

| route | How the two sides were built: nightly+patch, source-build or source-pair (see docs/OPERATIONS.md).

| changed lines covered | How many of the PR's changed, executable Python lines ran during the PR side's tests.

| re-parametrized | A test whose parameter ids changed (renamed cases); the new ids are judged on their own.

rebuilt from cleaned text

Written 2026-11-12 11:56 IST